

English translation: Batches 1–3

Problem formulation, analytical derivation, numerical analysis and discussion, and Appendix A

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Section numbers are retained. The two analytical queries identified in Batch 2 remain recorded in `AUTHOR_QUERIES_BATCH02.md`; Batch 3 uses the approved numerical implementation and does not treat numerical reproduction as an independent resolution of those analytical questions. The intended Figure-4 admissible-segments PDF is not currently committed; the wrapper therefore displays an explicit placeholder for that figure until the generated asset is restored.

2 Problem formulation

2.1 Geometry and local fracture configurations

Consider a piecewise homogeneous body under plane strain, consisting of two dissimilar isotropic elastic materials joined along a V-shaped interface. Material i has Young's modulus E_i and Poisson's ratio ν_i , where $i = 1, 2$ identifies the material. The angle between the two interface rays, measured through material 2, is 2α , where α is the half-angle and

$$0 < \alpha < \pi, \quad \alpha \neq \frac{\pi}{2}.$$

The value $\alpha = \pi/2$ corresponds to a straight interface and is treated only as a limiting case.

The origin of the polar coordinate system (r, θ) is placed at the interface corner O , where r is the distance from O and θ is measured from the bisector of the sector occupied by material 2. Symmetry about this bisector allows the analysis to be restricted to the half-plane $0 \leq \theta \leq \pi$. Material 2 occupies $0 < \theta < \alpha$, material 1 occupies $\alpha < \theta < \pi$, and the interface lies along $\theta = \alpha$. The radial and circumferential displacement components are denoted by u_r and u_θ , respectively. The circumferential normal stress is σ_θ , and the shear stress in polar coordinates is $\tau_{r\theta}$. On a radial line, these are the normal and tangential traction components, respectively.

The loading produces a local field that is symmetric about the bisector. For the intact interface, the singular part of the stress field along the interface is

$$\begin{aligned} \tau_{r\theta}(r, \alpha) &= Cg_1 r^{\lambda_0} + o(r^{\lambda_0}), \\ \sigma_\theta(r, \alpha) &= Cg_2 r^{\lambda_0} + o(r^{\lambda_0}), \quad r \rightarrow 0. \end{aligned}$$

Here, C is the generalized stress intensity factor incorporating the effects of the applied loading and the global body geometry; g_1 and g_2 are angular coefficients that depend on α and the elastic properties of the materials; and $\lambda_0 \in (-1, 0)$ is the singularity exponent of the intact corner. The notation $o(r^{\lambda_0})$ denotes a remainder whose ratio to r^{λ_0} tends to zero as $r \rightarrow 0$. The exponent λ_0 is the smallest real root of $D_0(-1 - \lambda_0) = 0$ in the open interval $(-1, 0)$ that is continuously connected to the physical solution branch. Here, D_0 is the characteristic determinant for the intact corner. Expressions for D_0 , g_1 , and g_2 are given in Appendix A.

When

$$Cg_2 < 0,$$

the normal stress along the interface is compressive. We consider a prescribed stationary V-shaped interfacial shear crack with two symmetric branches whose faces are in frictionless contact. Each branch has length l , measured from O along its interface ray, so the total crack length is $2l$. Consistent with the scope of the model, the applied loading is assumed insufficient to cause

loss of local stability at the outer crack tips. Branching at the corner and further interfacial propagation from the outer tips are competing mechanisms; the latter is not modeled here.

The corner of the cracked interface retains a power-law stress singularity [1]. This field can support a process zone in one of the materials, with subsequent formation of an opening-mode (Mode I) crack. Under the assumed symmetry of the geometry and loading, the bisector of the sector occupied by that material is a natural direction for the zone. Along this direction, the local in-plane shear-mode (Mode II) component at the zone tip vanishes by symmetry, consistent with the principle of local symmetry [2, 3]. Zones with arbitrary orientations are not considered. The bisector direction is therefore part of the assumed symmetric model and is not obtained by separately optimizing the branching direction.

Let β_i denote the polar angle of the bisector in material i :

$$\beta_1 = \pi, \quad \beta_2 = 0.$$

The subscript i on quantities associated with the process zone always identifies the material containing that zone.

The process zone in material i is modeled as a line of normal displacement discontinuity of length d_i , subject to a constant normal cohesive traction $\sigma_i > 0$, representing the material's resistance to separation. Figure 1 shows the zone in material 1 as a solid line and the alternative zone in material 2 as a dashed line. These are alternative configurations, considered separately.

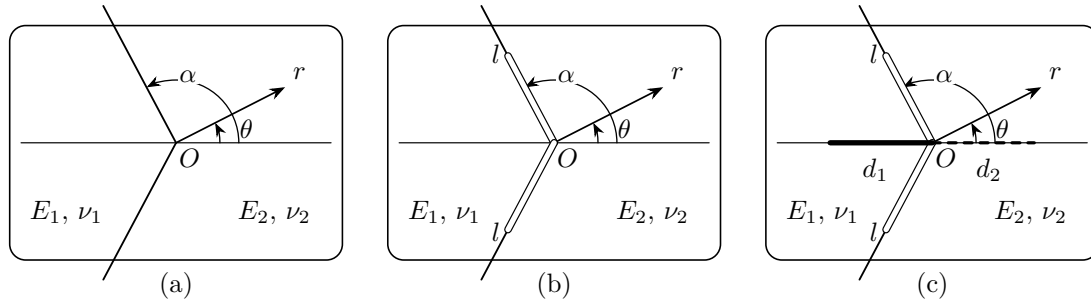


Figure 1: Local configurations near the interface corner: (a) intact V-shaped interface; (b) symmetric V-shaped interfacial shear crack with each branch of length l ; (c) process zone of length d_1 in material 1 and the alternative zone of length d_2 in material 2 (dashed line).

2.2 Local problem and boundary conditions

Let L be a characteristic outer dimension of the body. We assume the separation of scales

$$d_i \ll l \ll L$$

and neglect friction between the interfacial crack faces. In the local problem, the piecewise homogeneous body is replaced by an infinite plane, and the two interfacial crack branches are treated as semi-infinite. Their finite length and the applied loading enter through an asymptotic matching condition [4].

For a field quantity f defined on both sides of the ray $\theta = \alpha$, its jump across the interface is defined by

$$[[f]] = f(r, \alpha + 0) - f(r, \alpha - 0),$$

where $\alpha + 0$ and $\alpha - 0$ denote the limiting values approached from materials 1 and 2, respectively.

Symmetry, traction continuity, and frictionless contact give

$$\begin{aligned} \theta \in \{0, \pi\} : \quad \tau_{r\theta} &= 0, \\ \theta = \alpha : \quad \llbracket \sigma_\theta \rrbracket &= 0, \quad \llbracket \tau_{r\theta} \rrbracket = 0, \quad \tau_{r\theta} = 0, \quad \llbracket u_\theta \rrbracket = 0. \end{aligned} \quad (1)$$

The condition $\llbracket u_\theta \rrbracket = 0$ imposes normal contact between the crack faces, whereas $\tau_{r\theta} = 0$ expresses the absence of friction.

On the bisector of the material sector containing the process zone, the following mixed boundary conditions apply:

$$\begin{aligned} \theta = \beta_i, \quad 0 < r < d_i : \quad \sigma_\theta(r, \beta_i) &= \sigma_i, \\ \theta = \beta_i, \quad r > d_i : \quad u_\theta(r, \beta_i) &= 0, \\ \theta = \beta_j, \quad r > 0, \quad j \neq i : \quad u_\theta(r, \beta_j) &= 0. \end{aligned} \quad (2)$$

Here, $j \in \{1, 2\}$ identifies the other material. Thus, the symmetry condition on u_θ is replaced by a prescribed constant normal traction only over $0 < r < d_i$ along the active direction $\theta = \beta_i$.

At intermediate distances $d_i \ll r \ll l$, the solution must match the asymptotic field of the interfacial crack without a process zone [1]. In the inner problem, this matching condition is written as

$$\sigma_\theta(r, \beta_i) = CQ_i r^\lambda + o(r^{-1}), \quad r \rightarrow \infty. \quad (3)$$

Here, Q_i is the amplitude coefficient for material i , which depends on the interface angle, the elastic properties, and the branch length l , as specified in Appendix A. The exponent $\lambda \in (-1, 0)$ characterizes the corner singularity in the presence of the interfacial crack. It is the smallest real root of $D(-1 - \lambda) = 0$ in $(-1, 0)$ that is continuously connected to the physical branch, where D is the characteristic determinant for the cracked configuration. Expressions for D and Q_i are given in Appendix A. The limit $r \rightarrow \infty$ is understood in the inner-scale sense of asymptotic matching, rather than as the far field of the finite body; the remainder $o(r^{-1})$ decays faster than r^{-1} in that limit.

A process zone can form in material i only if the normal stress on its bisector is tensile. Equation (3) therefore gives the physical admissibility condition

$$CQ_i > 0.$$

The model thus requires both $Cg_2 < 0$, for contact between the interfacial crack faces, and $CQ_i > 0$, for formation of a process zone in material i .

2.3 Asymptotic fields near the process-zone tip

Near the outer tip of the displacement discontinuity, the stress and normal displacement gradient have the standard inverse-square-root asymptotic form for a homogeneous material:

$$\begin{aligned} \theta = \beta_i, \quad r \rightarrow d_i + 0 : \quad \sigma_\theta(r, \beta_i) &\sim \frac{k_i}{\sqrt{2\pi(r - d_i)}}, \\ \theta = \beta_i, \quad r \rightarrow d_i - 0 : \quad \frac{\partial u_\theta}{\partial r}(r, \beta_i) &\sim -\frac{2(1 - \nu_i^2)}{E_i} \frac{k_i}{\sqrt{2\pi(d_i - r)}}. \end{aligned} \quad (4)$$

Here, k_i is the local Mode I stress intensity factor at the process-zone tip; $r \rightarrow d_i + 0$ and $r \rightarrow d_i - 0$ denote approach from ahead of the zone and from within it, respectively. The

boundary-value problem is closed by imposing the stress-regularity condition

$$k_i = 0.$$

This condition removes the inverse-square-root singular term and is used below to determine the equilibrium zone length d_i . It does not require the bounded part of the stress at the zone tip to vanish.

3 Determination of the process-zone parameters

3.1 Mellin transform and functional equation

To construct the local solution, we introduce the dimensionless radial coordinate

$$\rho = \frac{r}{d_i}$$

and apply the Mellin integral transform to the boundary-value problem (1)–(3). Let p be the complex Mellin variable. The transforms of the normal stress outside the zone and the normal displacement gradient within the zone are defined, respectively, by

$$\begin{aligned}\Phi_i^+(p) &= \int_1^\infty \sigma_\theta(\rho d_i, \beta_i) \rho^p \, d\rho, \\ \Phi_i^-(p) &= \frac{E_i}{2(1-\nu_i^2)} \int_0^1 \frac{\partial u_\theta}{\partial r} \Big|_{\substack{r=\rho d_i \\ \theta=\beta_i}} \rho^p \, d\rho.\end{aligned}$$

The superscripts on Φ_i^+ and Φ_i^- identify functions assigned to the left and right half-planes, respectively; they do not indicate the sign of a physical quantity.

Eliminating the transforms of the interior field yields the functional equation

$$\Phi_i^+(p) + \frac{\sigma_i}{p+1} - \frac{CQ_i d_i^\lambda}{p+\lambda+1} = -\tan(\pi p) G_i(p) \Phi_i^-(p), \quad (5)$$

in the common strip of analyticity $-\varepsilon_2 < \operatorname{Re} p < \varepsilon_1$, where $\operatorname{Re} p$ denotes the real part of p and $\varepsilon_1, \varepsilon_2 > 0$ are sufficiently small constants. The kernel $G_i(p)$ is

$$G_i(p) = \frac{D_i(p) \cos(\pi p)}{D(p) \sin(\pi p)}.$$

Here, $D(p)$ is the characteristic determinant for the corner with an interfacial shear crack, given in Appendix A. For plane strain, define the dimensionless elastic ratio e and the material parameters κ_i by

$$e = \frac{1+\nu_2}{1+\nu_1} \frac{E_1}{E_2}, \quad \kappa_i = 3 - 4\nu_i.$$

The characteristic determinants $D_1(p)$ and $D_2(p)$ for configurations with a process zone in materials 1 and 2, respectively, are

$$\begin{aligned}D_1(p) &= 2e(1+\kappa_2)[\cos(2p\alpha) - \cos(2\alpha)]\{\sin^2[p(\pi-\alpha)] - p^2 \sin^2 \alpha\} \\ &\quad + (1+\kappa_1)[\sin(2p\alpha) + p \sin(2\alpha)][\sin(2p(\pi-\alpha)) - p \sin(2\alpha)], \\ D_2(p) &= 2(1+\kappa_1)[\cos(2p(\pi-\alpha)) - \cos(2\alpha)][\sin^2(p\alpha) - p^2 \sin^2 \alpha] \\ &\quad + e(1+\kappa_2)[\sin(2p(\pi-\alpha)) - p \sin(2\alpha)][\sin(2p\alpha) + p \sin(2\alpha)].\end{aligned}$$

In particular, the corner singularity exponent λ_i in the presence of a zone in material i is determined by

$$D_i(-1-\lambda_i) = 0,$$

where λ_i is the smallest real root in $(-1, 0)$ continuously connected to the corresponding physical branch. As before, $\alpha = \pi/2$ is understood only as a degenerate limit.

3.2 Wiener–Hopf factorization

We solve Eq. (5) by the Wiener–Hopf method [5], following the approach used for a related boundary-value problem [6]. The factors $G_i^\pm(p)$ are defined by the Cauchy integral

$$\exp \left\{ \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} \frac{\log G_i(z)}{z-p} dz \right\} = \begin{cases} G_i^+(p), & \text{Re } p < 0, \\ G_i^-(p), & \text{Re } p > 0, \end{cases}$$

where z is the complex integration variable and $i^2 = -1$. The contour runs upward along the imaginary axis. We also introduce the auxiliary gamma-function factors

$$K^\pm(p) = \frac{\Gamma(1 \mp p)}{\Gamma(\frac{1}{2} \mp p)}.$$

where Γ is the Euler gamma function. With the normalization $G_i^\pm(p) \rightarrow 1$ at infinity, the solution of the functional equation is

$$\begin{aligned} \Phi_i^+(p) &= -\frac{pG_i^+(p)}{K^+(p)} \left\{ \frac{\sigma_i}{p+1} \left[\frac{K^+(p)}{pG_i^+(p)} + \frac{K^+(-1)}{G_i^+(-1)} \right] \right. \\ &\quad \left. - \frac{CQ_id_i^\lambda}{p+\lambda+1} \left[\frac{K^+(p)}{pG_i^+(p)} + \frac{K^+(-\lambda-1)}{(\lambda+1)G_i^+(-\lambda-1)} \right] \right\}, \quad \text{Re } p < 0, \\ \Phi_i^-(p) &= K^-(p)G_i^-(p) \left\{ \frac{\sigma_i K^+(-1)}{(p+1)G_i^+(-1)} - \frac{CQ_id_i^\lambda K^+(-\lambda-1)}{(\lambda+1)(p+\lambda+1)G_i^+(-\lambda-1)} \right\}, \quad \text{Re } p > 0. \end{aligned} \tag{6}$$

For the kernels used below, $G_i(it)$ is positive and even in the real variable t , and tends to unity as $|t| \rightarrow \infty$. Thus, for real $p < 0$, the plus factor can be evaluated numerically using the real integral

$$\log G_i^+(p) = -\frac{p}{\pi} \int_0^\infty \frac{\log G_i(it)}{t^2 + p^2} dt, \quad p < 0.$$

In the numerical calculations, we check that the kernel is real and positive on the factorization contour and approaches unity at its far end.

3.3 Process-zone length

As $p \rightarrow +\infty$, the second relation in Eq. (6) gives

$$\Phi_i^-(p) \sim \frac{1}{\sqrt{p}} \left[\frac{\sigma_i K^+(-1)}{G_i^+(-1)} - \frac{CQ_id_i^\lambda K^+(-\lambda-1)}{(\lambda+1)G_i^+(-\lambda-1)} \right].$$

Applying an Abelian-type theorem [5] to the tip asymptotics (4) also gives

$$\Phi_i^-(p) \sim -\frac{k_i}{\sqrt{2pd_i}}.$$

Comparing these expressions yields the local stress intensity factor

$$k_i = -\sqrt{2d_i} \left[\frac{\sigma_i K^+(-1)}{G_i^+(-1)} - \frac{CQ_id_i^\lambda K^+(-\lambda-1)}{(\lambda+1)G_i^+(-\lambda-1)} \right].$$

Imposing $k_i = 0$, which removes the inverse-square-root singular term from the assumed tip asymptotics, gives the zone length:

$$d_i = \left[\frac{CQ_i K^+(-\lambda - 1) G_i^+(-1)}{\sigma_i (1 + \lambda) K^+(-1) G_i^+(-\lambda - 1)} \right]^{-1/\lambda}. \quad (7)$$

Since $-1 < \lambda < 0$, a real, positive d_i requires $CQ_i > 0$, consistent with the physical admissibility condition in Section 2.2. With geometry and material properties fixed, Eq. (7) also gives the power law $d_i \propto C^{-1/\lambda}$ on the admissible loading branch.

Alternatively, with the geometry, interfacial crack length, elastic properties, and applied loading fixed, the same equation gives the dependence on the normal cohesive traction:

$$d_i \propto \sigma_i^{1/\lambda} = \sigma_i^{-1/|\lambda|}.$$

Because $-1 < \lambda < 0$, decreasing σ_i increases the process-zone length.

Using the relation

$$Q_i = g_1 q_i l^{\lambda_0 - \lambda},$$

where g_1 is the intact-corner angular coefficient and q_i is a dimensionless amplitude coefficient, both given in Appendix A, we obtain from Eq. (7)

$$\frac{d_i}{l} = \left[\frac{Cl^{\lambda_0}}{\sigma_i} \frac{g_1 q_i K^+(-1 - \lambda) G_i^+(-1)}{(1 + \lambda) K^+(-1) G_i^+(-1 - \lambda)} \right]^{-1/\lambda}. \quad (8)$$

This relation also provides a check on the small-scale assumption $d_i/l \ll 1$ used in the local formulation.

3.4 Opening and energy parameter

Define the separation of the zone faces at the corner by $\delta_i = \llbracket u_\theta(0, \beta_i) \rrbracket$. The definition of $\Phi_i^-(p)$ gives

$$\delta_i = -2 \left(\frac{\partial u_\theta}{\partial r} \right)^* \Big|_{p=0, \theta=\beta_i} = -\frac{4(1 - \nu_i^2) d_i}{E_i} \Phi_i^-(0).$$

Here, the asterisk denotes the dimensional transform moment: for $f(r) = \partial u_\theta(r, \beta_i)/\partial r$, $f^*(p) = \int_0^{d_i} f(r)(r/d_i)^p dr$. The factor of two accounts for the symmetric displacement of the two zone faces. Substitution of Eq. (6), together with Eq. (7), gives

$$\delta_i = -\frac{4(1 - \nu_i^2) \sigma_i \lambda K^+(-1)}{E_i \sqrt{\pi G_i(0)} (1 + \lambda) G_i^+(-1)} d_i. \quad (9)$$

For fixed material properties and geometry, the coefficient multiplying d_i in Eq. (9) is constant. The opening therefore scales with the zone length as the applied loading varies: $\delta_i \propto d_i$. If the applied loading is fixed and σ_i is varied, the preceding power law instead gives

$$\delta_i \propto \sigma_i d_i \propto \sigma_i^{1+1/\lambda} = \sigma_i^{1-1/|\lambda|}.$$

The exponent $1 - 1/|\lambda|$ is negative. A decrease in the resistance to separation therefore increases both the zone length and its opening.

Within the adopted model, the work W_i done against the resistance to separation, per unit

out-of-plane thickness, is defined by [7]

$$W_i = \sigma_i \int_0^{d_i} \llbracket u_\theta(r, \beta_i) \rrbracket dr = -\frac{4(1 - \nu_i^2)\sigma_i}{E_i} d_i^2 \Phi_i^-(1).$$

Using Eq. (6), we obtain

$$W_i = -\frac{8\lambda(1 - \nu_i^2)\sigma_i^2}{\pi(\lambda + 2)[G_i^+(-1)]^2 E_i} d_i^2.$$

Differentiating with respect to the zone length along the family of equilibrium states, with the geometry and material properties fixed, defines the zone-related energy parameter J_i :

$$J_i = \frac{dW_i}{dd_i} = -\frac{16\lambda(1 - \nu_i^2)\sigma_i^2}{\pi(\lambda + 2)[G_i^+(-1)]^2 E_i} d_i. \quad (10)$$

For the numerical analysis, introduce the dimensionless loading, opening, and energy parameters σ' , δ'_i , and J'_i , respectively:

$$\sigma' = \frac{Cl^{\lambda_0}}{\sigma_i}, \quad \delta'_i = \frac{E_i \delta_i}{4(1 - \nu_i^2)\sigma_i l}, \quad J'_i = \frac{E_i J_i}{4(1 - \nu_i^2)\sigma_i^2 l}.$$

The primes here denote normalization, not differentiation. Equations (8), (9), and (10) then take the form

$$\begin{aligned} \frac{d_i}{l} &= \left[\sigma' \frac{g_1 q_i K^+(-1 - \lambda) G_i^+(-1)}{(1 + \lambda) K^+(-1) G_i^+(-1 - \lambda)} \right]^{-1/\lambda}, \\ \delta'_i &= -\frac{\lambda K^+(-1)}{\sqrt{\pi G_i(0)}(1 + \lambda) G_i^+(-1)} \frac{d_i}{l}, \\ J'_i &= -\frac{4\lambda}{\pi(\lambda + 2)[G_i^+(-1)]^2} \frac{d_i}{l}. \end{aligned}$$

All three relations use the corner exponent λ for the interfacial shear crack, i.e., the root of $D(-1 - \lambda) = 0$. The exponents λ_1 and λ_2 , which describe the corner singularity in the presence of a zone, do not replace λ in the formulas for the zone parameters.

Equating Eq. (10) to an independently specified critical value J_{ic} for material i provides a condition for the applied loading at which the process zone develops into a Mode I crack branching from the original interfacial crack. This criterion concerns local branching initiation within the adopted model.

4 Numerical analysis and discussion of results

The numerical analysis is performed for the normalized parameters d_i/l , δ'_i , and J'_i defined in Section 3. Only combinations of material, angle, and loading sign for which

$$Cg_2 < 0, \quad CQ_i > 0$$

hold simultaneously are regarded as physically admissible. The first inequality ensures compressive contact between the faces of the interfacial shear crack, whereas the second ensures tensile normal stress along the bisector of the angular sector of the material in which the process zone forms.

For the baseline material pair

$$E_1/E_2 = 0.5, \quad \nu_1 = \nu_2 = 0.3,$$

the sign changes of the coefficient g_2 occur at

$$\alpha_1 = 12.9202^\circ, \quad \alpha_2 = 107.0897^\circ.$$

These values are obtained by directly solving $D_0(-1 - \lambda_0) = 0$ together with $g_2 = 0$. For $C > 0$, a process zone is physically admissible in material 2 for $0 < \alpha < \alpha_1$ and in material 1 for $\alpha_2 < \alpha < 180^\circ$. For $C < 0$, the corresponding intervals are $\alpha_1 < \alpha < 90^\circ$ for material 1 and $90^\circ < \alpha < \alpha_2$ for material 2. The value $\alpha = 90^\circ$ corresponds to a straight interface and is treated below as a degenerate analytical limit rather than as an ordinary interior point of a numerical branch.

To assess the effect of elastic contrast, the same two transition angles were calculated for a range of $E_1/E_2 < 1$ with $\nu_1 = \nu_2 = 0.3$. The results are given in Table 1. As E_1/E_2 increases, the first transition angle α_1 increases monotonically, whereas α_2 decreases monotonically. Thus, elastic contrast systematically shifts the boundaries of the angular intervals of physical admissibility.

Table 1: Angles at which the coefficient g_2 changes sign for $\nu_1 = \nu_2 = 0.3$.

E_1/E_2	0.01	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90	0.99
$\alpha_1, ^\circ$	2.58	7.03	9.25	10.77	11.95	12.92	13.74	14.45	15.08	15.65	16.11
$\alpha_2, ^\circ$	108.02	107.90	107.72	107.52	107.30	107.09	106.88	106.69	106.50	106.32	106.17

4.1 Dependence on normalized loading

Figure 2 shows the process-zone parameters as functions of

$$\sigma' = \frac{Cl^{\lambda_0}}{\sigma_i}$$

for four reference configurations: $\alpha = 10^\circ$ ($i = 2, C > 0$), 45° ($i = 1, C < 0$), 105° ($i = 2, C < 0$), and 135° ($i = 1, C > 0$). For each configuration, only the physically admissible half-axis of the loading parameter is shown.

In all four cases, d_i/l , δ'_i , and J'_i increase monotonically with $|\sigma'|$. This synchronous behavior follows directly from the analytical relations $\delta'_i \propto d_i/l$ and $J'_i \propto d_i/l$ for fixed geometry. Among the four reference configurations shown, the largest values of all three parameters at $|\sigma'| = 0.5$

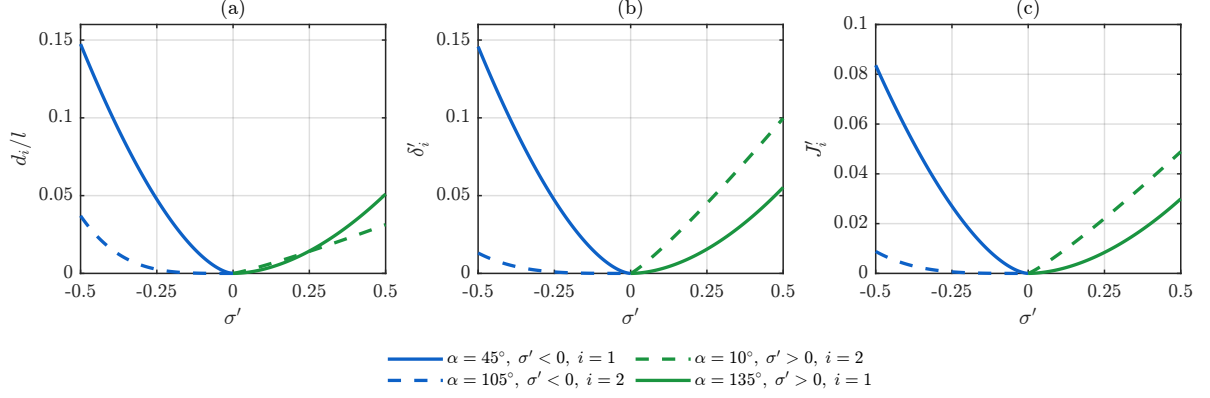


Figure 2: Dependence of the normalized process-zone parameters on the dimensionless loading parameter σ' for four reference angles and the physically admissible materials.

occur for $\alpha = 45^\circ$, $C < 0$; specifically,

$$\frac{d_1}{l} = 0.14743436, \quad \delta'_1 = 0.14544227, \quad J'_1 = 0.03716462.$$

This result is also relevant to the small-scale assumption: increasing the load simultaneously increases the branching-driving measures and moves the model closer to the limit of its asymptotic applicability.

For a fixed configuration, the opening δ_i and the zone-related energy parameter J_i are both uniquely linked to the equilibrium zone length d_i . Consequently, either quantity may be used as the basis of a deformation or energy criterion for branching initiation, provided that the corresponding critical material parameters are specified consistently. The present curves therefore describe two equivalent representations of the same one-parameter family of local equilibrium states rather than two independent evolution laws.

4.2 Dependence on the interface half-angle

Figure 3 shows the dependence of the normalized process-zone parameters on the half-angle α for the same baseline material pair at $\sigma' = \pm 0.5$. Only material/loading-sign combinations satisfying the physical-admissibility conditions are retained. Transitions between admissible branches occur at α_1 , 90° , and α_2 .

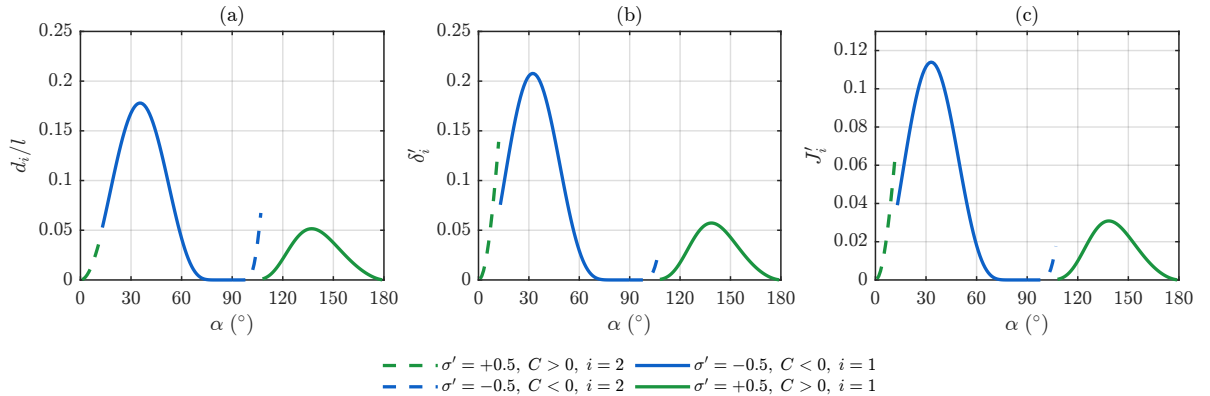


Figure 3: Dependence of the normalized process-zone parameters on the half-angle α at $\sigma' = \pm 0.5$. Only physically admissible combinations of material and loading sign are shown; $\alpha = 90^\circ$ is an analytical limit.

Within each principal admissible interval, the angular dependence has a pronounced maximum. Because δ'_i and J'_i are proportional to d_i/l at fixed α , the maxima of all three parameters are coincident. Physically, the broken-interface geometry affects not the material resistance threshold J_{ic} itself, but the value of the zone-related energetic driving measure attained at a prescribed external load. Hence, for a fixed J_{ic} , angles near a maximum correspond to less favorable geometrical configurations and a lower critical applied load.

The maximum of the blue branch in Figure 3 corresponds to d_i/l of order 0.18. This value is no longer unambiguously small compared with unity, whereas the local formulation is based on the assumption $d_i/l \ll 1$. The complete curve is therefore useful for examining the structure of the solution, but quantitative interpretation in the vicinity of the largest maximum should account for this limitation of the small-scale approximation.

Numerical evaluation of the plus factors for Figure 3 requires particular care as $\alpha \rightarrow 0$ and $\alpha \rightarrow 180^\circ$. A fixed factorization contour converges slowly near these limits. The calculations therefore use adaptive contour extension; a further increase in the tail changed the checked plus factors by at most 8.69×10^{-13} . Accordingly, Figure 3 is based on the converged adaptive values.

4.3 Singularity exponents and physical admissibility

For comparison, consider three local configurations: the intact bimaterial corner, the corner containing the prescribed interfacial shear crack, and the crack-plus-process-zone configuration. For the baseline material pair, their stress singularities are characterized by the exponents λ_0 , λ , and λ_i , respectively.

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Figure 4: Singularity exponents and physically admissible segments for the baseline material pair. Complete mathematical branches and physically admissible portions are distinguished according to the conditions $Cg_2 < 0$ and $CQ_i > 0$.

Within the physically admissible segments for the baseline material pair examined here, the configuration containing the interfacial shear crack has a substantially stronger singularity than the intact corner: $|\lambda|$ exceeds $|\lambda_0|$. Introducing the process zone partly reduces this concentration, and for the corresponding material

$$|\lambda_0| < |\lambda_i| < |\lambda|.$$

Thus, the process zone has a local shielding effect: it reduces the order of the singularity associated with the interfacial shear crack but does not eliminate the stress concentration completely. This relation is established here for the material pair considered and its physically admissible branches; it should not be extended to arbitrary elastic contrasts without additional verification.

The stronger singularity of the crack-containing configuration creates conditions favorable to local fracture in one of the materials, but the existence of a process zone is governed not

by the singularity exponent alone. It also depends on the conditions $Cg_2 < 0$ and $CQ_i > 0$, the separation resistance σ_i , and the level of external loading. This interpretation relates the sequence $\lambda_0 \rightarrow \lambda \rightarrow \lambda_i$ to the mechanical model for branching initiation without treating the singularity itself as a sufficient fracture criterion.

A Stress and displacement fields near the corner

This appendix collects the characteristic relations and asymptotic-field coefficients used in the main text. For plane strain, the dimensionless elastic ratio e and material parameters κ_i are

$$e = \frac{1 + \nu_2}{1 + \nu_1} \frac{E_1}{E_2}, \quad \kappa_i = 3 - 4\nu_i, \quad i = 1, 2.$$

For each characteristic exponent, the complex Mellin variable p and the generic singularity exponent λ_* are related by

$$p = -1 - \lambda_*, \quad -1 < \lambda_* < 0.$$

The physical exponent is taken to be the smallest real root in $(-1, 0)$ that is continuously connected to the corresponding solution branch. The case $\alpha = \pi/2$ is the degenerate straight-interface limit and is not treated as an ordinary interior point of a characteristic branch.

A.1 Intact corner

For the intact interface, the singular part of the interfacial stress field is [1]

$$\begin{aligned} \tau_{r\theta}(r, \alpha) &= C g_1 r^{\lambda_0} + o(r^{\lambda_0}), \\ \sigma_\theta(r, \alpha) &= C g_2 r^{\lambda_0} + o(r^{\lambda_0}), \end{aligned} \quad r \rightarrow 0, \quad (\text{A.1})$$

where C is the generalized stress intensity factor, λ_0 is the intact-corner singularity exponent, and the dimensionless angular coefficients g_1 and g_2 are

$$\begin{aligned} g_1 &= (1 - e) \sin[\lambda_0(\pi - \alpha)] [(\lambda_0 + 1) \sin 2\alpha + \sin\{2(\lambda_0 + 1)\alpha\}] \\ &\quad + e(1 + \kappa_2) \sin(\lambda_0\pi) \sin[(\lambda_0 + 2)\alpha], \\ g_2 &= (1 - e) \cos[\lambda_0(\pi - \alpha)] [(\lambda_0 + 1) \sin 2\alpha + \sin\{2(\lambda_0 + 1)\alpha\}] \\ &\quad + e(1 + \kappa_2) \sin(\lambda_0\pi) \cos[(\lambda_0 + 2)\alpha]. \end{aligned}$$

The exponent λ_0 is determined by the characteristic equation

$$D_0(-1 - \lambda_0) = 0, \quad -1 < \lambda_0 < 0, \quad (\text{A.2})$$

where the intact-corner determinant $D_0(p)$ is

$$D_0(p) = b_0(p) + e b_1(p) + e^2 b_2(p),$$

with auxiliary functions $b_0(p)$, $b_1(p)$, and $b_2(p)$ given by

$$\begin{aligned} b_0(p) &= [\sin(2p\alpha) + p \sin(2\alpha)] [\kappa_1 \sin\{2p(\pi - \alpha)\} + p \sin(2\alpha)], \\ b_1(p) &= (1 + \kappa_1)(1 + \kappa_2) \sin^2(p\pi) \\ &\quad - [\sin(2p\alpha) + p \sin(2\alpha)] [\kappa_1 \sin\{2p(\pi - \alpha)\} + p \sin(2\alpha)] \\ &\quad - [\sin\{2p(\pi - \alpha)\} - p \sin(2\alpha)] [\kappa_2 \sin(2p\alpha) - p \sin(2\alpha)], \\ b_2(p) &= [\sin\{2p(\pi - \alpha)\} - p \sin(2\alpha)] [\kappa_2 \sin(2p\alpha) - p \sin(2\alpha)]. \end{aligned}$$

A.2 Corner with an interfacial shear crack

For the prescribed symmetric interfacial shear crack, each branch has length l . At distances $r \ll l$, the normal stress along the bisector of the sector occupied by material i has the asymptotic form [1]

$$\sigma_\theta(r, \beta_i) = CQ_i r^\lambda + o(r^\lambda), \quad r \rightarrow 0, \quad \beta_1 = \pi, \quad \beta_2 = 0. \quad (\text{A.3})$$

Here, λ is the corner singularity exponent for the cracked configuration and β_i is the corresponding bisector direction. The amplitude coefficient Q_i is expressed as

$$Q_i = g_1 q_i l^{\lambda_0 - \lambda},$$

where the dimensionless coefficient q_i and the derivative $D'(p)$ of the cracked-corner determinant are

$$q_i = \frac{2S_i(-1-\lambda)K^+(-1-\lambda)G^+(-1-\lambda)}{(\lambda-\lambda_0)D'(-1-\lambda)K^+(-1-\lambda_0)G^+(-1-\lambda_0)}, \quad D'(p) = \frac{dD(p)}{dp}.$$

The functions $S_i(p)$ are auxiliary numerator functions defined below. The factors $K^+(p)$ and $G^+(p)$ are specified later in this subsection; G^+ belongs to the base kernel G , distinct from the zone kernel G_i in Section 3. Explicitly,

$$\begin{aligned} S_1(p) &= 2e(1+\kappa_2)s_{11}(p)\sin[(p+1)\alpha] \\ &\quad + s_{12}(p)\left[(1+\kappa_1)\sin\{p(\pi-\alpha)-\alpha\} + 2(1-e)s_{13}(p)\right], \\ S_2(p) &= s_{21}(p)\left[e(1+\kappa_2)\sin[(p+1)\alpha] + 2(1-e)s_{22}(p)\right] \\ &\quad - 2(1+\kappa_1)\sin\{p(\pi-\alpha)-\alpha\}s_{23}(p), \end{aligned}$$

where the auxiliary functions $s_{mn}(p)$, with $m = 1, 2$ and $n = 1, 2, 3$, are

$$\begin{aligned} s_{11}(p) &= p\cos(p\pi-\alpha)\sin\alpha - \sin[p(\pi-\alpha)]\cos(p\alpha), \\ s_{12}(p) &= p\sin 2\alpha + \sin(2p\alpha), \\ s_{13}(p) &= p\cos[p(\pi-\alpha)]\sin\alpha - \sin[p(\pi-\alpha)]\cos\alpha, \\ s_{21}(p) &= \sin[2p(\pi-\alpha)] - p\sin 2\alpha, \\ s_{22}(p) &= p\cos(p\alpha)\sin\alpha + \sin(p\alpha)\cos\alpha, \\ s_{23}(p) &= p\cos(p\pi+\alpha)\sin\alpha + \cos[p(\pi-\alpha)]\sin(p\alpha). \end{aligned}$$

The corner singularity exponent λ for the configuration containing the interfacial crack satisfies

$$D(-1-\lambda) = 0, \quad -1 < \lambda < 0, \quad (\text{A.4})$$

where the characteristic determinant $D(p)$ is

$$D(p) = a_0(p) + e a_1(p),$$

with auxiliary functions $a_0(p)$ and $a_1(p)$ given by

$$\begin{aligned} a_0(p) &= (1+\kappa_1)[\cos\{2p(\pi-\alpha)\} - \cos 2\alpha][\sin(2p\alpha) + p\sin 2\alpha], \\ a_1(p) &= (1+\kappa_2)[\cos(2p\alpha) - \cos 2\alpha][\sin\{2p(\pi-\alpha)\} - p\sin 2\alpha]. \end{aligned}$$

The coefficients q_i use the base kernel

$$G(p) = \frac{(e + \kappa_1)(1 + e\kappa_2) \sin(\pi p) D(p)}{(e + \kappa_1 + 1 + e\kappa_2) \cos(\pi p) D_0(p)}$$

and its plus factor

$$G^+(p) = \exp \left\{ \frac{1}{2\pi i} \int_{-i\infty}^{i\infty} \frac{\log G(z)}{z - p} dz \right\}, \quad \text{Re } p < 0.$$

Here, z is the complex integration variable, $i^2 = -1$, and the contour runs upward along the imaginary axis. The gamma-function factor is

$$K^+(p) = \frac{\Gamma(1 - p)}{\Gamma(\frac{1}{2} - p)}.$$

where Γ is the Euler gamma function. For real $p < 0$, the verified numerical algorithm evaluates $G^+(p)$ equivalently using the real integral

$$\log G^+(p) = -\frac{p}{\pi} \int_0^\infty \frac{\log G(it)}{t^2 + p^2} dt.$$

where t is a real integration variable.

A.3 Corner with a process zone

For a process zone in material i , the kernel of the functional equation in Section 3 is

$$G_i(p) = \frac{D_i(p) \cos(\pi p)}{D(p) \sin(\pi p)}.$$

The corresponding characteristic determinants are

$$\begin{aligned} D_1(p) &= 2e(1 + \kappa_2)[\cos(2p\alpha) - \cos 2\alpha]\{\sin^2[p(\pi - \alpha)] - p^2 \sin^2 \alpha\} \\ &\quad + (1 + \kappa_1)[\sin(2p\alpha) + p \sin 2\alpha][\sin\{2p(\pi - \alpha)\} - p \sin 2\alpha], \\ D_2(p) &= 2(1 + \kappa_1)[\cos\{2p(\pi - \alpha)\} - \cos 2\alpha][\sin^2(p\alpha) - p^2 \sin^2 \alpha] \\ &\quad + e(1 + \kappa_2)[\sin\{2p(\pi - \alpha)\} - p \sin 2\alpha][\sin(2p\alpha) + p \sin 2\alpha]. \end{aligned} \tag{A.5}$$

The exponent λ_i for a configuration with a zone in material i is determined by

$$D_i(-1 - \lambda_i) = 0, \quad -1 < \lambda_i < 0, \quad i = 1, 2. \tag{A.6}$$

These two roots are used for the λ_1 and λ_2 curves in Figure 4. They characterize the corner singularity in the presence of a process zone and must not replace λ in the formulas for d_i , δ_i , and J_i , where λ is defined by Eq. (A.4).

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